

Programs and Tools: Installations, Tips and Tricks

Andrés Sicard-Ramírez

October 25, 2025

Some instructions for installing and using some programs and tools.

Contents

1	Beamer	1
2	Debian/GNU Linux on Windows via WSL	2
3	Emacs	3
4	Frama-C	3
5	GHC	4
6	ghcup	5
7	Lamba Shell	5
8	L^AT_EX	6

1 Beamer

Removing the headline and footline from the title page

Last modification: 2025-04-12

Tested with Beamer 3.72.

Use the `plain` option for removing the headline and footline areas from the title page:

```
\begin{frame}[plain]
\titlepage
\end{frame}
```

Removing the navigation symbols

Last modification: 2025-04-12

Tested with Beamer 3.72.

Write in the preamble the command

```
\setbeamertemplate{navigation symbols}{{}}
```

for suppressing all the navigation symbols.

2 Debian/GNU Linux on Windows via WSL

Last modification: 2025-04-12

For installing Debian on Windows we shall use the WSL (Windows Subsystem for Linux).

(i) Running PowerShell

For installing WSL open PowerShell in administrator mode by right-clicking and selecting *Run as administrator*:

```
Windows PowerShell  
C:\WINDOWS\system32>
```

(ii) Installing WSL

For installing WSL use the command

```
C:\WINDOWS\system32> wsl --install  
Downloading: Windows Subsystem for Linux 2.4.13  
Installing: Windows Subsystem for Linux 2.4.13  
...  
The operation completed successfully.
```

(iii) Available Linux distributions

To see a list of available Linux distributions available for installing with WSL use the command

```
C:\WINDOWS\system32> wsl --list --online
```

(iv) Installing Debian

For installing Debian run the commands

```
C:\WINDOWS\system32> wsl --install -d Debian
C:\WINDOWS\system32> wsl -d Debian
Enter new UNIX name:
New password:
Retype new password:
```

(v) Updating Debian

After installing Debian you should update it by running the commands:

```
$ sudo apt update
$ sudo apt upgrade
```

3 Emacs

Fixing trailing whitespace issues

Last modification: 2025-04-12

Tested with Emacs 29.4.

Add to the `.emacs` file the instruction `(setq-default show-trailing-whitespace t)` for showing whitespace and run the command `M-x delete-trailing-whitespace` for removing them.

4 Frama-C

Last modification: 2025-10-25

Tested with Framac-C 31.0

For installing the program Frama-C we assume you are using Debian or WSL/Debian (see Section 2).

First, run the following commands:

```
$ opam switch create frama-c-31.0 4.14.1  
$ eval $(opam env --switch=frama-c-31.0)  
$ opam install frama-c  
  
<> frama-c.31.0 installed successfully ><><><><><><><><><><><>  
=> The Frama-C/WP plug-in requires one or more external prover(s).  
    Recommended provers are:  
      - Alt-Ergo (https://alt-ergo.ocamlpro.com)  
      - CVC4     (https://cvc4.github.io)
```

```
- CVC5      (https://cvc5.github.io)
- Z3        (https://github.com/Z3Prover/z3)
Use 'why3 config detect' to configure new provers.
```

```
=> Ivette is a new GUI for Frama-C, currently in development.
Run 'ivette' once to finalize installation (requires an internet connection).
Once finalized, 'ivette' will work offline.
Finalization also requires Node v20 or v22 and Yarn:
- install NVM (https://github.com/nvm-sh/nvm)
- run 'nvm use 22'
- run 'npm install --global yarn'
```

Now, run the following command:

```
Found prover Alt-Ergo version 2.6.2, OK.
Found prover Alt-Ergo version 2.6.2 (alternative: counterexamples)
Found prover Alt-Ergo version 2.6.2 (alternative: BV)
Found prover Coq version 8.19.2, but no Why3 libraries were compiled for it
3 prover(s) added
Save config to /home/asr/.why3.conf
```

5 GHC

Last modification: 2025-08-27

For installing GHC we suggest to use the `ghcup` tool (see Section 6). If after the installation you need a different version of GHC, you can also use the `ghcup` tool. For example,

(i) Installing GHC 9.12.2

```
$ ghcup install ghc 9.12.2
```

(ii) Switching to GHC 9.12.2

```
$ ghcup set ghc 9.12.2
```

(iii) Testing the current version of GHC

```
$ ghc --version
The Glorious Glasgow Haskell Compilation System, version 9.12.2
```

6 ghcup

Last modification: 2025-08-27

For installing the `ghcup` tool we assume you are using Debian or WSL/Debian (see Section 2).

- (i) Before installing `ghcup` you need to install the required [Debian packages](#) for your version of Debian:

```
$ cat /etc/debian_version
12.10
```

```
$ sudo apt install build-essential curl libffi-dev libffi8 \
libgmp-dev libgmp10 libncurses-dev libncurses5 libtinfo5
```

- (ii) Now, [run the following command](#) for installing `ghcup` and chose the default options:

```
$ curl --proto '=https' --tlsv1.2 -sSf https://get-ghcup.haskell.org | sh
```

- (iii) Restart your terminal for the changes to take effect.

- (iv) Testing your installation by running the command

```
$ ghcup --version
The GHCup Haskell installer, version 0.1.50.1
```

- (v) Testing the GHC version installed by default.

```
$ ghc --version
The Glorious Glasgow Haskell Compilation System, version 9.6.7
```

7 Lamba Shell

Last modification: 2025-08-27

The Lambda Shell “is a feature-rich shell environment and command-line tool for evaluating terms of the pure, untyped lambda calculus” writing in Haskell.

The next instructions for installing the Lamba Shell program assume you are using Debian or WSL/Debian (see Section 2) and you installed GHC via the `ghcup` tool (see Section 6).

Instructions for installing the Lamba Shell program:

- (i) Install GHC 9.12.2, if necessary (see Section 5).

- (ii) Installing `git`, if necessary

```
$ sudo apt install git
```

- (iii) Cloning the GitHub repository

```
$ git clone https://github.com/asr/lambda-shell.git  
$ cd lambda-shell
```

- (iv) Installing Lambda Shell

```
$ cabal install
```

- (v) Testing your installation

```
$ lambdaShell --version  
  
The Lambda Shell, version 0.9.9.1  
Copyright 2005-2011, Robert Dockins
```

8 L^AT_EX

Obsolete commands

Last modification: 2025-04-12

Do not use `\bf`, `\em`, `\it`, `\rm`, `\sf`, `\sl` or `\tt`.

The above commands have some issues. For example, how produce bold italic text?

Both `{\bf \it Hello World!}` or `{\it \bf Hello World!}` do not work. You should use `\textbf{\textit{Hello World!}}` or `\textit{\textbf{Hello World!}}` instead.

See, [What's wrong with `\bf`, `\it`, etc.?](#)